SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:		**Date:**	05/10/26

Code of Conduct

I M. Srilekha / 1919260007 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the Student: M. Srilekha

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Q1. Two-Room Vacuum World

Given

Rooms: A and B

Agent starts at A

Initially: A = Dirty, B = Dirty

Actions: Suck, Right, Left, NoOp

Total time steps: T = 6

Clean square = +1

Left/Right movement penalty =

(a) PEAS Description — 4 Marks

PEAS	Answer
P – Performance	+1 for every clean square at the end of each step; −1 for every Left/Right movement
E – Environment	Two-room world: A and B
A – Actuators	Suck, Left, Right, NoOp
S – Sensors	Detects its current location and whether the square is Dirty/Clean

Interpretation: The agent should keep both rooms clean while avoiding unnecessary movement.

(b) Agent X — 5 Marks

Agent X rule

If current square is Dirty → Suck
If Clean → move to the other square
Starting condition:
A = Dirty, B = Dirty

Step	Location	Percept	Action	Clean squares	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	−1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	Left	2	−1	1
5	A	Clean	Right	2	−1	1
6	B	Clean	Left	2	−1	1

Total score

$$1+0+2+1+1+1=\boxed{6}$$

So,
Agent X total score = 6

(c) Agent Y —4 Marks

Agent Y hasinternal state. Once it remembers that both rooms are clean, it chooses instead of continuing to move.

Step	Location	Percept	Action	Clean squares	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	−1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	NoOp	2	0	2
5	B	Clean	NoOp	2	0	2
6	B	Clean	NoOp	2	0	2

Total score

$$1+0+2+2+2+2=\boxed{9}$$

So,
Agent Y total score = 9

(d) Score as a function of p —3 Marks

For Agent X:
There are4 movements.
Clean-square points:

$$1+1+2+2+2+2=10$$

Therefore,

$$\boxed{X=10-4p}$$

For Agent Y:
There is only1 movement.
Clean-square points:

$$1+1+2+2+2+2=10$$

Therefore,

$$\boxed{Y=10-p}$$

When is Y better?

$$10-p>10-4p \; -p>-4p \; 3p>0$$

Therefore,

$$p > 0$$

So Agent Y is strictly better than Agent X whenever the movement penalty is positive.

(e) Which Agent is Rational? —4 Marks

Answer:

Agent Y is rational

Reason:The performance measure rewards clean squares but gives a penalty for unnecessary movement. Agent Y remembers that both squares are already clean and chooses NoOp, avoiding unnecessary movement.

Agent X cannot do this because it hasno internal state. When it sees a clean square, its rule always tells it to move to the other square.

Therefore, even though X and Y receive the same percepts, Y uses its percept history/ internal state to make a better decision.

Final interpretation:

Agent Y achieves a higher score because it considers what happened previously and stops when further movement is unnecessary. This makes Y the rational agent under the given performance measure.

Q2. Task Environment Analysis and Agent Selection

There are 3 environments:

E1: Autonomous taxi in city traffic

E2: Robot picking defective parts froma moving conveyor belt

E3: Chess-playing agent with a clock

The six properties are Observability, Number of agents, Determinism, Episodic/Sequential, Static/Dynamic, Discrete/Continuous.

(a) Classify all environments —9 Marks

Property	E1: Taxi	E2: Part-picking Robot	E3: Chess
Observability	Partially observable	Partially observable	Fully observable
Agents	Multi-agent	Single agent	Multi-agent
Determinism	Stochastic	Stochastic	Deterministic
Episodic/Sequential	Sequential	Sequential	Sequential
Static/Dynamic	Dynamic	Dynamic	Semi-dynamic
Discrete/	Continuous	Continuous	Discrete

ne-phrase justification

E1 – Taxi

Partially observable → Cannot know everything about traffic/pedestrians.

Multi-agent → Other vehicles and pedestrians act independently.

Stochastic → Traffic events are uncertain.

Sequential → Current action affects future driving.

Dynamic → Traffic changes while the taxi is operating.

Continuous → Position, speed and time vary continuously.

E2 – Robot

Partially observable → Camera may not give complete information.

Single agent → The robot performs the picking task.

Stochastic → Defect detection and moving parts involve uncertainty.

Sequential → Each action affects the next action.

Dynamic → Conveyor belt keeps moving.

Continuous → Position and movement are continuous.

E3 – Chess

Fully observable → Entire chessboard is visible.

Multi-agent → Two players compete.

Deterministic → A legal move has a predictable board result.

Sequential → Each move affects future moves.

Semi-dynamic → Board does not change by itself, but the clock changes.

Discrete → Board positions and moves are discrete.

The paper specifically defines the harder values as1 point, easier values as0, and semi-dynamic as0.5 for Static/Dynamic. CSA17-AI_CO1_AT1_Analytical_Pro…

(b) Difficulty Score —4 Marks

Rule

Harder property =

Easier property =0

Semi-dynamic =0.5

E1 – Taxi

Partially observable = 1

Multi-agent = 1

Stochastic = 1

Sequential = 1

Dynamic = 1

Continuous = 1

$$\boxed{E1=6}$$

E2 – Robot

Partially observable = 1

Single agent = 0

Stochastic = 1

Sequential = 1

Dynamic = 1

Continuous = 1

$$\boxed{E2=5}$$

E3 – Chess

Fully observable = 0

Multi-agent = 1

Deterministic = 0

Sequential = 1

Semi-dynamic = 0.5

Discrete = 0

$$\boxed{E3=2.5}$$

Ranking

$$\boxed{E1 > E2 > E3}$$

Hardest → easiest:

E1 Taxi → E2 Robot → E3 Chess

(c) Agent Architecture —4 Marks

Hardest: E1 Taxi

$$\boxed{\text{Utility-based agent}}$$

Why?
The taxi has partial observability, multiple agents, uncertainty, continuous movement and changing traffic. It must choose the best action considering different possible outcomes.

Easiest: E3 Chess

$$\boxed{\text{Goal-based agent}}$$

Why?
Chess is fully observable and deterministic, with a clear goal: win the game/checkmate the opponent

(d) Interpretation —3 Marks

Theenvironment properties determine how complex the agent program needs to be

For example, a taxi needs a more complex agent because the environment is observable, stochastic, multi-agent, dynamic and continuous

Chess is comparatively simpler because it is fully observable, deterministic and discrete although it is still sequential and multi-agent.

Final sentence to write:

Therefore, the complexity of an AI agent depends not only on the application domain but also on the properties of its task environment.

Question 3: Turing Test Data Across the History of AI

Given:

System	Interrogations	Correctly Identified
ELIZA	50	47
Chatbot B	50	36
System C	50	29

(a) Correct-identification rate and fooling rate

Formula:

Correct-identification rate
= (Correctly identified / Total interrogations) × 100

Fooling rate
= (Total −Correctly identified) / Total × 100

ELIZA:

Correct-identification rate
= (47/50) × 100 = 94%

Fooling rate
= (50−47)/50 × 100 = 6%

Chatbot B:

Correct-identification rate
= (36/50) × 100 = 72%

Fooling rate
= (50−36)/50 × 100 = 28%

SystemC:

Correct-identification rate
= (29/50) × 100 = 58%

Fooling rate
= (50−29)/50 × 100 = 42%

Answer:

System	Correct-identification rate	Fooling rate
ELIZA	94%	6%
Chatbot B	72%	28%
System C	58%	42%

(b) Turing's 70%criterion

Turing's criterion ₹0%

ELIZA:

94% > 70% → Does not satisfy

Distance from 50% chance level:

94 − 50 = 44 percentage points

Chatbot B:

72% > 70% → Does not satisfy

Distance from 50% chance level:

72 − 50 = 22 percentage points

System C:

58% < 70% → Satisfies

Distance from 50% chance level:

58 − 50 = 8 percentage points

Final answer:
Only System C satisfies Turing's 70% criterion

(c) Improvement in fooling rate

Fooling rates:

ELIZA = 6%
Chatbot B = 28%
System C = 42%

ELIZA → System C:

Percentage-point improvement:

42 − 6 = 36 percentage points

Ratio:

42/6 = 7

Therefore, System C's fooling rate 7 times ELIZA's.

Chatbot B → System C:

42 − 28 = 14 percentage points

(d) Interpretation

The Turing Test measures acting humanly

System C has a fooling rate of 42%, meaning the machine was taken as human in 42%

of the interrogations.

A high fooling rate does not prove that SystemC is actually thinking or understanding.

According to theChinese Room argument, producing human-like answers does not necessarily mean that the systemunderstands their meaning.

Conclusion:
A high fooling rate shows human-like behaviour, but it does not prove true understanding, thinking or rationality.

Question 4: Rational Route Choice under Uncertainty

Given:

Deadline =55 minutes

Utility formula:

[
U = 100 - T - 20L
]

Where,

(T) = travel time in minutes

(L=1) if the taxi arrives after 55 minutes

(L=0) otherwise

Route	Normal Time	Probability of Disruption	Disrupted Time
R1: Highway	30 min	0.30	70 min
R2: City road	45 min	0.10	60 min
R3: Ring road	50 min	0	50 min

(a) Utility of each route —4 Marks

Formula:

[
U=100-T-20L
]

R1 – Highway

Normal:(T=30,\ L=0)

[
U=100-30-20(0)
]

[
\boxed{U=70}
]

Disrupted:(T=70,\ L=1) → Late

[
U=100-70-20(1)
]

[
\boxed{U=10}
]

R2 – City road

Normal:(T=45,\ L=0)

[
U=100-45-20(0)
]

[
\boxed{U=55}
]

Disrupted:(T=60,\ L=1) → Late

[
U=100-60-20(1)
]

[
\boxed{U=20}
]

R3 – Ring road

Normal:(T=50,\ L=0)

$$
U=100-50-20(0)
$$

$$
\boxed{U=50}
$$

Disrupted:No disruption.

$$
\boxed{U=50}
$$

Final answer:

Route	Normal Utility	Disrupted Utility
R1	70	10 (Late)
R2	55	